

MANGALORE REFINERY - INSPECTION DEPARTMENT

Pressure Vessel Inspection Report No. INSP/PV/2025/0417

Equipment Tag: V-2104 (Crude Unit Desalter Drum) Unit: CDU-II
Date of Inspection: 14-08-2025 Inspector: R. Shenoy, Sr. Engineer (Insp.)
Design Pressure: 18.5 kg/cm²(g) Design Temp: 150 C MOC: SA-516 Gr.70
Nominal Shell Thickness: 22.0 mm Minimum Required Thickness: 18.4 mm

FINDINGS

1. Ultrasonic thickness at bottom shell CML-07 measured 19.1 mm, a loss of 2.9 mm.
2. Corrosion rate at CML-07 is 0.48 mm per year; remaining life about 1.5 years.
3. Pitting up to 1.2 mm depth observed near the brine outlet nozzle N4.
4. External coating damaged over approximately 3 square metres on the south side.
5. Pressure safety valve PSV-2104A test certificate expired on 30-06-2025.
6. No leaks, bulging or cracks were observed on welds or nozzles.

RECOMMENDATIONS

- a) Re-test and re-certify PSV-2104A before the unit restarts.
- b) Repair coating on the south side within 30 days.
- c) Monitor CML-07 every 6 months; plan shell repair in the next turnaround.

Remarks: Vessel fit for continued service
till next TA subject to PSV re-certification.
Nozzle N4 pitting - recheck in Dec 2025.

- R. Shenoy 14/8/25

Approved by: _____
Head (Inspection)